

Notes: Heyerdahl-Larsen & Illeditsch (JFE, 2026)

Demand Disagreement

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1 Big picture

- **Question.** Why do investors disagree about future interest rates even after accounting for disagreement about macroeconomic fundamentals such as inflation, consumption, GDP, unemployment, profits, and housing starts?
- **Main puzzle.** Standard heterogeneous-belief asset-pricing models usually predict a tight connection between disagreement about fundamentals and disagreement about asset prices. The data show that this connection is incomplete. The authors call this the **disagreement correlation puzzle**.
- **Related puzzle.** Traditional asset-pricing models also imply that prices and returns should be strongly tied to fundamentals. The low empirical correlation between asset prices/returns and fundamental variables is the classic **correlation puzzle**.
- **Core idea.** Investors can disagree not only about future fundamentals, but also about future **demand for savings**. This demand disagreement comes from uncertainty and disagreement about the future fraction of patient versus impatient investors in the economy.
- **Mechanism.** Patient and impatient investors have different time discount rates. They also hold biased beliefs about how common their type will be in the future. These beliefs create speculative trade in long-duration assets and move prices even when aggregate output fundamentals are known.
- **Model environment.** The paper builds a continuous-time overlapping-generations exchange economy with log utility, two preference types, heterogeneous beliefs about demand composition, and two Brownian shocks: an output/supply shock and a demand-composition shock.
- **Empirical measure.** Demand disagreement is proxied by the component of Survey of Professional Forecasters yield disagreement that is orthogonal to disagreement about macroeconomic fundamentals.
- **Main empirical result.** Demand disagreement explains a large residual portion of yield disagreement. It is positively related to bond yields and yield volatility and predicts future bond excess returns.
- **Main model result.** Demand disagreement generates stochastic yield volatility, time-varying bond risk premia, an upward-sloping yield curve, and a low correlation between asset prices and fundamentals.
- **Economic takeaway.** Asset prices can move because investors disagree about *who future marginal investors will be*, not only because they disagree about future output or inflation.

2 Data and measurement

2.1 Survey of Professional Forecasters data

- The empirical analysis uses quarterly data from the Survey of Professional Forecasters (SPF).
- The sample covers the third quarter of 1981 through the second quarter of 2024.
- Forecasters report forecasts for the three-month Treasury bill rate over the next four quarters.
- The SPF also contains forecasts for macroeconomic variables, including CPI, real GDP, aggregate consumption, federal government consumption and investment, corporate profits after tax, residential fixed investment, nonresidential fixed investment, unemployment, corporate bond yields, and housing starts.
- Yield disagreement is measured as the cross-sectional standard deviation of yield forecasts across forecasters.

Interpretation. The SPF is useful because it allows the authors to observe, at the same time, disagreement about yields and disagreement about macroeconomic fundamentals. This makes it possible to ask whether yield disagreement is simply macro-disagreement in another form.

2.2 Correlation puzzle data

- To motivate the classic correlation puzzle, the paper reports correlations between stock returns, trading volume, and one-year real yields with dividend growth and consumption growth.
- The sample uses Shiller’s and NYSE webpage data and covers yearly observations from 1891 to 2009.
- The correlations are low at one-, five-, and ten-year horizons, especially between returns or yields and consumption growth.

Interpretation. If fundamentals fully drove prices, returns and yields should be strongly related to consumption and dividend growth. Table 1 shows that this is not generally true.

2.3 Measuring demand disagreement

For each forecast horizon $\Delta t \in \{1Q, 2Q, 3Q, 4Q\}$, the authors run a pooled forecast-level regression:

$$\hat{y}_{t,t+\Delta t}^i = \beta_0 + \beta_X \hat{X}_{t,t+\Delta t}^{i,F} + \beta_y \hat{y}_{t,t}^i + \varepsilon_{t,t+\Delta t}^i. \quad (1)$$

Here $\hat{y}_{t,t+\Delta t}^i$ is forecaster i ’s Treasury yield forecast, $\hat{X}_{t,t+\Delta t}^{i,F}$ is the vector of macroeconomic forecasts, and $\hat{y}_{t,t}^i$ is the forecaster’s nowcast of the yield.

Demand disagreement is then defined as the cross-sectional standard deviation of the residual:

$$DD_{t,t+\Delta t} = SD_i(\varepsilon_{t,t+\Delta t}^i). \quad (2)$$

Interpretation. The residual isolates the part of yield disagreement that is not explained by disagreement about standard macroeconomic fundamentals. The authors interpret this residual disagreement as disagreement about future demand for savings and consumption.

2.4 What demand disagreement predicts

- It rises with the forecast horizon.
- It accounts for a meaningful share of observed T-bill disagreement.
- It is positively related to nominal yield volatility at one- to five-year maturities.
- It predicts future excess returns on nominal discount bonds.
- It is useful even after controlling for yields and macroeconomic disagreement in Internet Appendix tests.

Interpretation. The empirical results make demand disagreement a priced state variable rather than a purely survey-based curiosity.

3 Model and theoretical structure

3.1 Aggregate output and overlapping generations

Aggregate output follows

$$dY_t = \mu_Y Y_t dt + \sigma_Y Y_t dZ_{Y,t}, \quad (3)$$

where $Z_{Y,t}$ is the output or supply shock.

Agents live for a random time. Death arrives with hazard rate $\nu > 0$, and a continuum of newborn agents enters every instant. This keeps total population size constant and creates an overlapping-generations structure.

Interpretation. OLG structure matters because newborn investors can have different patience and different beliefs, so demand composition changes over time even without changes in output fundamentals.

3.2 Patient and impatient investors

There are two preference types:

- patient investors a have a low time discount rate ρ^a ;
- impatient investors b have a high time discount rate $\rho^b > \rho^a$.

The fraction of patient newborns is

$$\alpha_t = \frac{1}{1 + \exp(-l_t)}, \quad (4)$$

where l_t follows the Ornstein–Uhlenbeck process

$$dl_t = \kappa(\bar{l} - l_t)dt + \sigma_l dZ_{\alpha,t}. \quad (5)$$

Interpretation. A positive demand-composition shock raises the fraction of patient agents. Since patient investors want to save more, this changes the demand for long-duration assets.

3.3 Beliefs and false consensus bias

Investors observe the current state l_t but disagree about its long-run mean. Type $i \in \{a, b\}$ believes

$$dl_t = \kappa(\bar{l}^i - l_t)dt + \sigma_l dZ_{\alpha,t}^i. \quad (6)$$

Patient investors are optimistic that patient types will be more common in the future, while impatient investors are pessimistic:

$$\bar{l}^a \geq \bar{l} \geq \bar{l}^b. \quad (7)$$

Interpretation. The paper interprets this as a false-consensus-style bias: each type believes its own preference type is more prevalent than it truly is.

3.4 Stochastic discount factor

The equilibrium stochastic discount factor has the form

$$\xi_t = \frac{X_t}{Y_t}, \quad (8)$$

where X_t captures variation in state prices due to belief and time-preference heterogeneity.

Its dynamics can be written as

$$\frac{d\xi_t}{\xi_t} = -r_t dt - \theta_{Y,t} dZ_{Y,t} - \theta_{\alpha,t} dZ_{\alpha,t}. \quad (9)$$

Interpretation. Standard models price only output risk. Here there is also a price of demand risk, $\theta_{\alpha,t}$, because the future mix of patient and impatient investors affects equilibrium asset prices.

3.5 Risk-free rate and price of demand risk

The real short rate is

$$r_t = E_f[\rho] + \mu_Y - \sigma_Y^2 + \nu(1 - \alpha_t\beta_t^a - (1 - \alpha_t)\beta_t^b), \quad (10)$$

where

$$E_f[\rho] = f_t\rho^a + (1 - f_t)\rho^b \quad (11)$$

is the consumption-share-weighted discount rate.

The price of output risk is

$$\theta_{Y,t} = \sigma_Y, \quad (12)$$

and the price of demand shock risk is

$$\theta_{\alpha,t} = \sigma_\Delta \left(\frac{1}{2} - f_t \right). \quad (13)$$

Interpretation. Demand risk is priced positively when impatient investors consume more than patient investors ($f_t < 1/2$) and negatively when patient investors dominate ($f_t > 1/2$). At $f_t = 1/2$, the market view aligns with the statistical view and demand risk is not priced.

3.6 Wealth-consumption ratio

The equilibrium wealth-consumption ratio is affine in the consumption share of patient investors:

$$\phi_t = \phi^a f_t + \phi^b (1 - f_t), \quad (14)$$

where

$$\phi^a = \frac{1}{\rho^a + \nu}, \quad \phi^b = \frac{1}{\rho^b + \nu}, \quad \phi^a > \phi^b. \quad (15)$$

Interpretation. A higher patient consumption share raises aggregate wealth relative to consumption because patient agents discount future payoffs less.

3.7 Consumption-share dynamics

The endogenous state variable f_t evolves as

$$df_t = \mu_{f,t}dt + \sigma_{f,t}dZ_{\alpha,t}, \quad (16)$$

with

$$\sigma_{f,t} = f_t(1 - f_t)\sigma_\Delta. \quad (17)$$

Interpretation. Disagreement makes the consumption share volatile. The volatility is zero at the boundaries and highest near the middle, where patient and impatient investors both have wealth with which to speculate.

3.8 Consol and stock market

The price of the consol equals the wealth-consumption ratio:

$$B_t^C = \phi_t. \quad (18)$$

The stock return has both cash-flow exposure and consol-price exposure:

$$d \log P_{s,t} = d \log Y_{s,t} + d \log B_t^C. \quad (19)$$

Interpretation. The stock is exposed to output shocks through cash flows and to demand shocks through the valuation ratio. Bonds and stocks therefore both inherit demand-disagreement effects.

4 Empirical and model design

4.1 How the paper tests the mechanism

- First, it documents that disagreement about macroeconomic fundamentals explains only part of disagreement about future interest rates.
- Second, it constructs demand disagreement as the residual part of yield disagreement after conditioning on macro-forecast disagreement.
- Third, it checks whether this residual disagreement behaves like the model’s demand-disagreement state variable.
- Fourth, it compares model-implied yield volatility, bond risk premia, real yields, and fitted demand disagreement to empirical counterparts.

Interpretation. This is not a causal identification paper. It is a theory-plus-measurement paper: the empirical tests ask whether a model-implied state variable has the predicted asset-pricing patterns.

4.2 Goodness-of-fit exercise

- The model has two latent state variables: the newborn patient fraction α_t and the consumption share of patient investors f_t .
- The authors estimate these states using an Unscented Kalman Filter.
- The observed series used for filtering are demand disagreement and the two-year TIPS yield.
- Longer-maturity TIPS yields and nominal bond yields are left out of the filtering and used for validation.

Interpretation. The filtering exercise asks whether a parsimonious model with one exogenous and one endogenous state can reproduce multiple bond-market moments without directly fitting them all.

4.3 What would reject the model

- If residual yield disagreement had no relation to yields, volatility, or bond excess returns, the demand-disagreement interpretation would be weak.
- If the model could only fit moments by directly fitting all of them, the mechanism would be less convincing.
- If demand disagreement predicted yield volatility but not risk premia, the pricing-channel interpretation would be incomplete.

Interpretation. The paper’s empirical case is strongest because the same measure is connected to both volatility and expected return patterns.

5 Tables and figures

This section preserves the compact visual-notes format. Adjacent figures and tables are grouped to save space, and all images are directly cropped from the original paper.

5.1 Table 1 + Table 2 + Figure 1: the correlation and disagreement correlation puzzles

Correlation puzzle. This table presents the correlations between dividend and consumption growth with (i) stock returns, (ii) percentage changes in trading volume, and (iii) changes in the one-year real yield. The data are sourced from Shiller's and NYSE web pages and cover yearly observations from 1891 to 2009. The results indicate a low correlation between asset returns/trading volume and fundamental factors such as dividends and consumption growth.

	Stock returns		Trading volume		One-year yield	
	Dividends	Consumption	Dividends	Consumption	Dividends	Consumption
1 year	0.08	0.00	0.21	0.29	0.26	-0.16
5 year	0.46	0.28	0.35	0.38	0.20	0.04
10 year	0.65	0.11	0.31	0.40	0.24	0.06

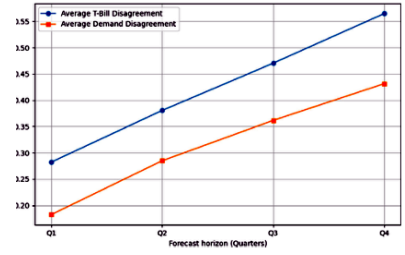
ata, output risk, and uncertainty is distinct in that 9.9 Disagreement correlation puzzle

(a) Table 1: correlation puzzle.

Table 2
Disagreement correlation puzzle. This table presents the unadjusted R^2 values from regressions of the disagreement in the T-bill rate on fundamental disagreements for one to four quarters ahead. The fundamental variables include CPI, real GDP (RGDP), aggregate consumption (RCONSUM), federal government consumption and gross investments (RFEDGOV), corporate profits after tax (CPROF), residential fixed investment (RRESINV), nonresidential fixed investments (RNRESINV), unemployment (UNEMP), corporate bond yields (BOND), and housing starts (HOUSING). The regressions are conducted using the level, changes, and logarithmic transformations of these variables. The results indicate that fundamental disagreements explain only a modest portion of the variation in T-bill rate disagreements.

Quarters	Level	Changes	Log
1	0.719	0.318	0.497
2	0.769	0.224	0.553
3	0.765	0.217	0.604
4	0.743	0.240	0.627

(b) Table 2: disagreement correlation puzzle.



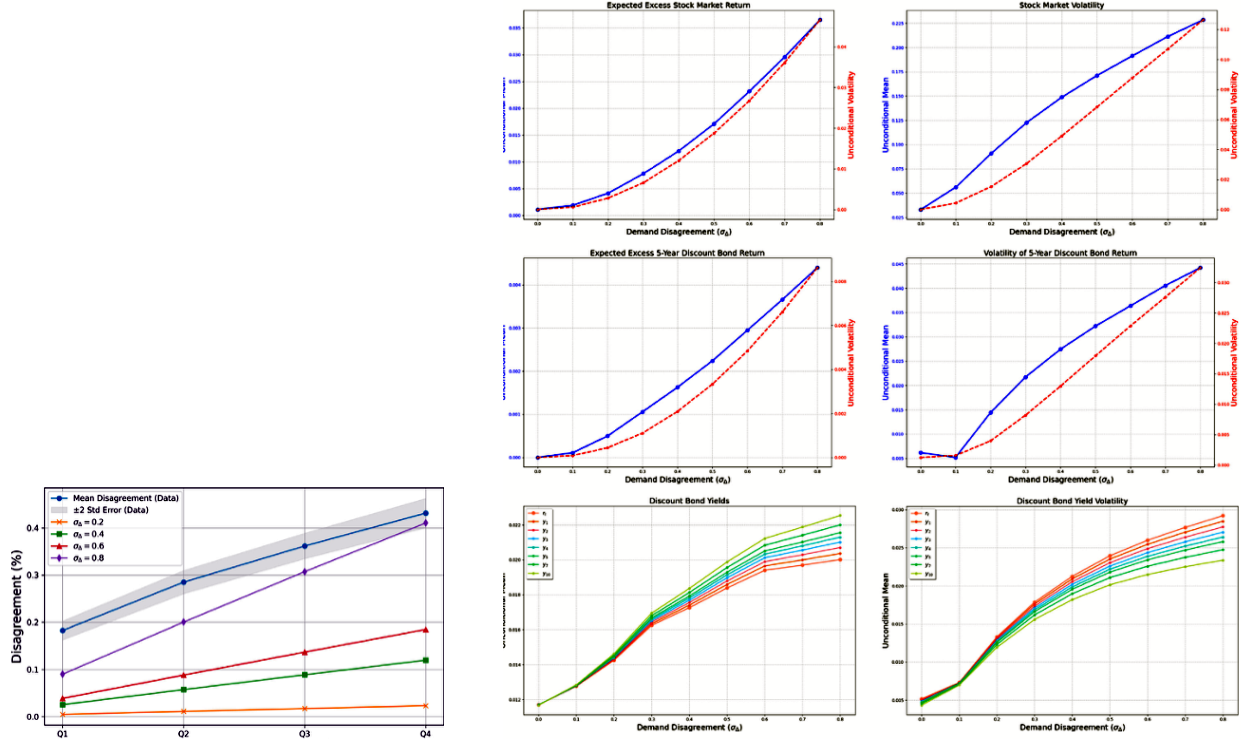
(c) Figure 1: measuring demand disagreement.

Read:

- Table 1 shows low correlations between asset returns/trading volume/real yields and consumption or dividend growth.
- Stock-return correlations with consumption growth are especially low: 0.00 at one year, 0.28 at five years, and 0.11 at ten years.
- Table 2 shows that fundamental disagreement explains only part of T-bill disagreement. Depending on transformation and horizon, R^2 values range from about 0.217 to 0.769.
- Figure 1 shows that the residual demand-disagreement component rises with the forecast horizon and accounts for a large fraction of yield disagreement.

Economic message. Prices and yield forecasts are not fully pinned down by macro fundamentals. A distinct demand-side disagreement component remains and motivates the model.

5.2 Figure 2 + Figure 3: model-implied disagreement and unconditional asset-pricing moments



(a) Figure 2: yield disagreement due to demand disagreement.

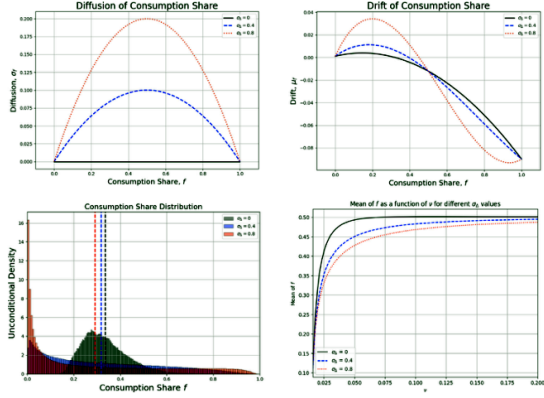
(b) Figure 3: unconditional asset-pricing moments.

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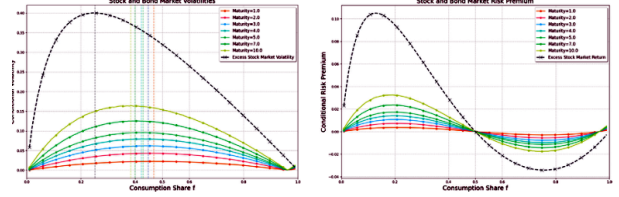
- Figure 2 compares model-implied yield disagreement with data. Higher σ_Δ produces more yield disagreement at every forecast horizon.
- The baseline calibration uses $\sigma_\Delta = 0.8$ because it generates disagreement levels close to the lower side of the data.
- Figure 3 shows that increasing demand disagreement raises expected stock returns, stock-market volatility, five-year bond excess returns, five-year bond return volatility, yields, and yield volatility.
- The bottom-left panel shows a steeper yield curve as disagreement increases.

Economic message. Demand disagreement is not just a belief variable. In equilibrium, it changes valuation ratios, volatility, and risk premia.

5.3 Figure 4 + Figure 5: consumption-share dynamics and conditional risk premia



(a) Figure 4: consumption share.



(b) Figure 5: conditional return volatility and risk premium.

Read:

- Figure 4 shows that consumption-share volatility is hump-shaped in f_t and increases with demand disagreement.
- The distribution of f_t becomes more dispersed when σ_Δ rises, raising the probability of extreme consumption-share realizations.
- The unconditional mean consumption share of patient investors falls as disagreement rises because speculative trading shifts wealth toward impatient investors in parts of the state space.
- Figure 5 shows that bond and stock return volatility are highest when both investor groups have material consumption shares.
- Conditional bond risk premia are positive when impatient investors dominate and negative when patient investors dominate.

Economic message. The key state variable is not output but the distribution of wealth and consumption across preference types. Demand disagreement affects prices by shifting who becomes marginal in asset markets.

5.4 Tables 3–4: yield volatility and bond risk premium regressions

Table 3

Yield volatility regression coefficients. This table presents the standardized regression coefficients of nominal yield volatility for discount bonds with maturities from one to five years, regressed on yield disagreement due to demand disagreement as implied by both the model and the data. In the model, yield volatility is estimated using a simulation of 500,000 years of monthly data, regressing actual conditional volatility on yield disagreement based on one-year real yield forecasts. In the empirical analysis, nominal yield volatility is estimated using an AR(1)-GARCH(1,1) model and regressed on the demand disagreement measure constructed in Section 2.3. The model's standardized coefficients (population values) are consistent with those observed in the data. Newey–West corrected standard errors with four lags are used, and t-statistics for the data are reported in brackets. This table demonstrates that there is a positive relation between yield volatility and demand disagreement in the model and the data.

Maturity	1	2	3	4	5
Data	0.7334 (14.108)	0.7284 (13.902)	0.7204 (13.584)	0.6978 (12.738)	0.6618 (11.544)
Model	0.9938	0.9937	0.9928	0.9913	0.9892

(a) Table 3: yield volatility regression coefficients.

Table 4

Bond risk premium regression coefficients. This table presents the standardized regression coefficients of excess bond returns for discount bonds with maturities, T , ranging from two to five years, regressed on yield disagreement due to demand disagreement as implied by both the model and the data. In the model, we use a simulation of 500,000 years of data to regress the instantaneous bond risk premium, $\lambda_{R^T,t}$, on yield disagreement based on one-year real yield forecasts, as all yield disagreement in the model arises from demand disagreement. In the empirical analysis, bond risk premia are calculated as $rx_{t,t+4}^{(T)} = T y_t^{(T)} - (T-4)y_{t+4}^{(T-4)} - y_t^{(1)}$, where each period is a quarter, and are regressed on our demand disagreement measure constructed in Section 2.3. The standardized coefficients from both the model (population values) and the data are reported. Newey–West corrected standard errors with four lags are used, and t-statistics for the empirical data are reported in brackets. This table demonstrates that an increase (decrease) in demand disagreement increases (decreases) future excess bond returns.

Maturity	2	3	4	5
Data	0.4364 (5.845)	0.4085 (5.361)	0.3932 (5.084)	0.3837 (4.922)
Model	0.1510	0.1505	0.1494	0.1478

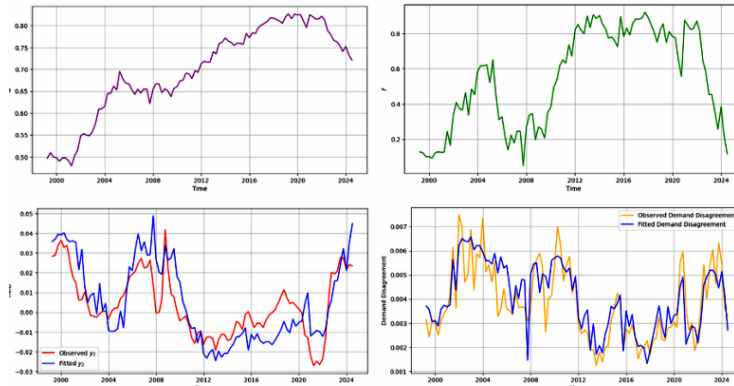
(b) Table 4: bond risk premium regression coefficients.

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- Table 3 shows positive standardized coefficients of demand disagreement for nominal yield volatility across one- to five-year maturities.
- In the data, coefficients decline mildly from 0.7334 at one year to 0.6618 at five years; all reported t-statistics are large.
- The model produces coefficients close to one because yield disagreement is the sole driver of model yield volatility in the simulation.
- Table 4 shows positive coefficients for future bond risk premia at two- to five-year maturities.
- Empirical coefficients are economically meaningful: approximately 0.4364, 0.4085, 0.3932, and 0.3837 for maturities two through five years.

Economic message. Demand disagreement predicts both volatility and returns. This is important because it shows a risk-premium channel, not merely noise in survey forecasts.

5.5 Figure 6 + Table 5: goodness of model fit



ing the goodness-of-model-fit. The figure shows the estimated σ_t in the top-left, the estimated f_t in the top-right, and the observed

(a) Figure 6: estimated states and fitted observables.

Table 5
Regression results: Yields, yield volatilities, and bond risk premia. The table displays regression results of observed data against model-implied values based on filtered x_t and f_t . Coefficients are estimated from the regression $y_t^{obs} = a + b y_t^{mod} + \epsilon_t$. Dashes indicate missing data for certain maturities. Although coefficients differ from one, their statistical significance and correct signs demonstrate that the model captures essential data features.

Maturity	Real yields			Real yield volatilities			Nominal bond risk premia		
	Coeff.	Std. Dev.	R ²	Coeff.	Std. Dev.	R ²	Coeff.	Std. Dev.	R ²
2 years	0.621	0.062	0.667	0.141	0.095	0.035	1.354	0.543	0.178
3 years	0.593	0.057	0.694	0.156	0.073	0.086	1.540	0.672	0.143
5 years	0.558	0.054	0.687	0.136	0.045	0.155	1.283	0.708	0.084
7 years	0.535	0.053	0.663	0.111	0.038	0.136	–	–	–
10 years	0.517	0.052	0.629	0.079	0.032	0.088	–	–	–
Slope	1.147	0.275	0.238	–	–	–	–	–	–

fect bond yields. The results of this alternative approach are far to those reported here.

model. Through these extensions, we demonstrate that while line model is intentionally simplistic, its core insights remain within more generalized frameworks. For brevity, we re-

(b) Table 5: yields, volatilities, and bond risk premia.

Read:

- Figure 6 shows the filtered patient newborn fraction α_t , the patient consumption share f_t , the observed and fitted two-year real yield, and the observed and fitted demand disagreement.
- The model tracks demand disagreement and the two-year real yield reasonably well using only a small number of state variables.
- Table 5 compares observed data with model-implied values for real yields, real-yield volatilities, and nominal bond risk premia.
- The model explains the level and slope of real yields well; R^2 values for real yields are around 0.63–0.69 across maturities.
- The model also captures the cyclicalities of bond risk premia, though the fit is weaker than for yields.

Economic message. A parsimonious two-state-variable demand-disagreement model can explain multiple bond-market moments that are difficult to reconcile in standard fundamental-disagreement models.

6 Short summary

- The paper studies disagreement about future interest rates and shows that disagreement about macro fundamentals explains only part of it.
- It calls this the disagreement correlation puzzle: asset-price disagreement is not tightly linked to fundamental disagreement, contrary to many heterogeneous-belief models.
- The authors introduce demand disagreement: disagreement about the future distribution of patient and impatient investors.
- They construct a proxy for demand disagreement using the residual component of SPF yield forecasts after controlling for macro-forecast disagreement.
- In the model, patient and impatient investors have different discount rates and disagree about how prevalent their types will be in the future.
- The consumption share of patient investors is the key endogenous state variable.
- Demand disagreement creates speculative trade and makes the price of long-duration assets move independently of output fundamentals.
- The price of demand risk is positive when impatient investors dominate and negative when patient investors dominate.
- Demand disagreement raises yield volatility, bond return volatility, and expected bond excess returns.
- The model produces an upward-sloping yield curve and time-varying risk premia.
- Empirically, demand disagreement is positively related to yield volatility and predicts future bond excess returns.
- A Kalman-filtered version of the model fits real yields and demand disagreement reasonably well.
- The paper’s main insight is that asset prices can reflect uncertainty about future marginal investors, not only uncertainty about future cash flows or macro fundamentals.

7 Conclusion

7.1 Core conclusion

Heyerdahl-Larsen and Illeditsch argue that disagreement about future demand for savings can explain why investors disagree about future interest rates even after accounting for macroeconomic disagreement. Their model resolves the disagreement correlation puzzle by adding a non-fundamental but economically meaningful source of asset-price disagreement: uncertainty about the future composition of patient and impatient investors.

7.2 Economic intuition

When impatient investors dominate consumption, long-duration bonds appear underpriced from the econometrician's perspective and command a positive demand-risk premium. When patient investors dominate, the same bonds appear overpriced and demand-risk premia can turn negative. Because disagreement amplifies both exposure to demand shocks and the price of those shocks, unconditional bond premia and yield volatility rise with demand disagreement.

7.3 Takeaway

The paper expands the disagreement literature beyond beliefs about inflation, output, or dividends. It shows that beliefs about who future investors will be can move prices, generate risk premia, and break the tight link between fundamentals and asset returns. For asset pricing, demand disagreement is a separate state variable.